Dear Hiring Team,

I am applying for the AI Software Test Engineer position at Dassault Systèmes. My primary engineering motivation is understanding the physics and mathematics behind a model and translating that understanding into a computational representation. Your role's focus on the credibility of FEA datasets and physics-based surrogate predictions connects directly to that motivation and to my simulation research.

At the University of Texas at San Antonio, I developed a neural-network surrogate trained on high-fidelity XFEM simulations to approximate localized damage behavior in larger structural models. That work connects the two kinds of models described in your posting: a numerical representation of physical behavior and a learned approximation built from its results. The relevant contribution I would bring is experience working at that interface. It provides a foundation for examining how simulation assumptions affect training data and for interpreting a surrogate's output in the context of the physical problem it represents.

My research also included creating a structural digital twin of an experimental habitat. I acquired impact-hammer and accelerometer data, identified natural frequencies and mode shapes, and calibrated a parameterized beam-element model against measured modal properties and acceleration histories. This provides complementary experience with checking a computational representation against independent measurements. For a role concerned with dataset fidelity and validation criteria, that background offers a practical basis for investigating discrepancies and communicating what a model comparison establishes, as well as the limits of the evidence supporting it.

At L&T Technology Services, I developed Jenkins CI/CD pipelines with Bash and PowerShell scripting to streamline debugging and test workflows. I would bring that implementation experience alongside my simulation background while learning your 3DEXPERIENCE and cloud-HPC workflows. I would welcome a discussion about how these capabilities could support reliable simulation data and surrogate-model validation within SIMULIA.

Sincerely,
Adnan Shahriar